Extra Dessert #19: Exam Blueprint and Score Tracker

What the Extra exam looks like, and a one-page grid to track your readiness across all 50 question groups.

System organizer · Use first; revisit weekly during practice · Self-study reference · Practice exams at eham.net/exams

The Extra Exam At A Glance

Fact	Value
Pool source	NCVEC 2024-2028 Amateur Extra Class question pool
Pool size	599 questions, 50 exam groups, 10 subelements (E0-E9)
Exam length	50 questions
Questions per group on exam	One random question per exam group, drawn from that group's published pool
Pass threshold	37 of 50 correct (74%)
Time limit	Usually liberal – no fast-paced clock pressure
Format	Multiple choice, four options (A, B, C, D)

Subelement Breakdown (Questions Per Subelement On The Exam)

The Extra pool is divided into 10 subelements. Each subelement contributes a fixed number of questions to your exam.

Sub-element	Topic	Qs on exam
E1	Commission's rules	6
E2	Operating procedures	5
E3	Radio wave propagation	3
E4	Amateur practices	5
E5	Electrical principles	4
E6	Circuit components	6
E7	Practical circuits	8
E8	Signals and emissions	4
E9	Antennas and transmission lines	8
E0	Safety	1
Total	**All subelements**	**50**

Pass-grade implication: losing all 7 questions from the **reserved-for-self-study** groups (E5B, E5C, E6C, E7G, E8C, E8D, E9G) still leaves 43 covered exam questions. Hitting 86% on those covered questions = 37 correct = pass.

Score Tracker (One Row Per Exam Group)

Mark each box as you progress. **Covered** = the Bite/Course/Dessert that owns this group. **Read** = you have read the handout. **Mock OK** = you got the question right on at least one practice exam. **Weak** = flag for review.

Reserved-for-self-study groups (R) are listed at the bottom; expect to guess these on exam day unless you have done outside study.

Grp	Owner	Topic	Read	Done	Mock	Weak
E0A	D5	Safety, RF exposure, grounding	â˜	â˜	â˜	â˜
E1A	D12	Frequency privileges and band edges	â˜	â˜	â˜	â˜
E1B	D13	Station restrictions, RACES	â˜	â˜	â˜	â˜
E1C	D6	Auto/remote control, foreign ops	â˜	â˜	â˜	â˜
E1D	D7	Space and Earth stations, telemetry	â˜	â˜	â˜	â˜
E1E	D14	VE program, exam admin	â˜	â˜	â˜	â˜
E1F	D15	Misc rules: amps, spread spectrum, STA	â˜	â˜	â˜	â˜
E2A	D8	Satellite operations	â˜	â˜	â˜	â˜
E2B	D9	Television practices (ATV, SSTV)	â˜	â˜	â˜	â˜
E2C	D10	DX, contest, remote operating	â˜	â˜	â˜	â˜
E2D	D16	VHF/UHF digital, APRS, EME	â˜	â˜	â˜	â˜
E2E	D3	HF digital modes	â˜	â˜	â˜	â˜
E3A	D1	Specialized propagation	â˜	â˜	â˜	â˜
E3B	D4	Propagation effects	â˜	â˜	â˜	â˜
E3C	D11	Propagation prediction, space weather	â˜	â˜	â˜	â˜
E4A	D17	Test equipment	â˜	â˜	â˜	â˜
E4B	EC-B	VNA, frequency counter, power meter	â˜	â˜	â˜	â˜
E4C	EC-B	Receiver performance, MDS	â˜	â˜	â˜	â˜
E4D	B13	Dynamic range, intermod	â˜	â˜	â˜	â˜
E4E	D2	Noise and interference	â˜	â˜	â˜	â˜
E5A	EC-A	Resonance, Q, bandwidth	â˜	â˜	â˜	â˜
E5D	B5	Skin effect, reactive power	â˜	â˜	â˜	â˜
E6A	B3	Semiconductor device recognition	â˜	â˜	â˜	â˜
E6B	B10	Diodes	â˜	â˜	â˜	â˜
E6D	B11	Inductors, transformers, piezo	â˜	â˜	â˜	â˜
E6E	B4	MMICs, RF packaging	â˜	â˜	â˜	â˜
E6F	B12	Electro-optical devices	â˜	â˜	â˜	â˜
E7A	B14	Digital logic, truth tables	â˜	â˜	â˜	â˜
E7B	B6	Amplifier classes, stages	â˜	â˜	â˜	â˜
E7C	B17	Filters, matching networks	â˜	â˜	â˜	â˜
E7D	B1	Power supplies, regulators	â˜	â˜	â˜	â˜
E7E	B15	Modulators, mixers, detectors	â˜	â˜	â˜	â˜
E7F	B7	SDR and DSP basics	â˜	â˜	â˜	â˜
E7H	B8	Oscillators, PLL, DDS	â˜	â˜	â˜	â˜
E8A	EC-D	Fourier, RMS, PEP, ADC	â˜	â˜	â˜	â˜
E8B	EC-D	Modulation index, FDM/TDM/OFDM	â˜	â˜	â˜	â˜
E9A	EC-C	Antenna basics, ERP/EIRP, ground	â˜	â˜	â˜	â˜
E9B	B16	Antenna patterns, Method of Moments	â˜	â˜	â˜	â˜
E9C	B2	Wire antennas, phased arrays	â˜	â˜	â˜	â˜
E9D	B9	Yagis, parabolic, short antennas	â˜	â˜	â˜	â˜
E9E	B18	Impedance matching, phasing lines	â˜	â˜	â˜	â˜
E9F	EC-C	Transmission lines, velocity factor	â˜	â˜	â˜	â˜
E9H	D18	Receiving antennas, RDF, Beverages	â˜	â˜	â˜	â˜
E5B	**R**	**Phase angles, time constants, susceptance**	â˜	â˜	â˜	â˜
E5C	**R**	**Complex impedance, phasors**	â˜	â˜	â˜	â˜
E6C	**R**	**Digital ICs, gate families**	â˜	â˜	â˜	â˜
E7G	**R**	**Operational amplifiers**	â˜	â˜	â˜	â˜
E8C	**R**	**Digital signals, QAM**	â˜	â˜	â˜	â˜
E8D	**R**	**Keying defects, spread spectrum, codes**	â˜	â˜	â˜	â˜
E9G	**R**	**Smith chart**	â˜	â˜	â˜	â˜